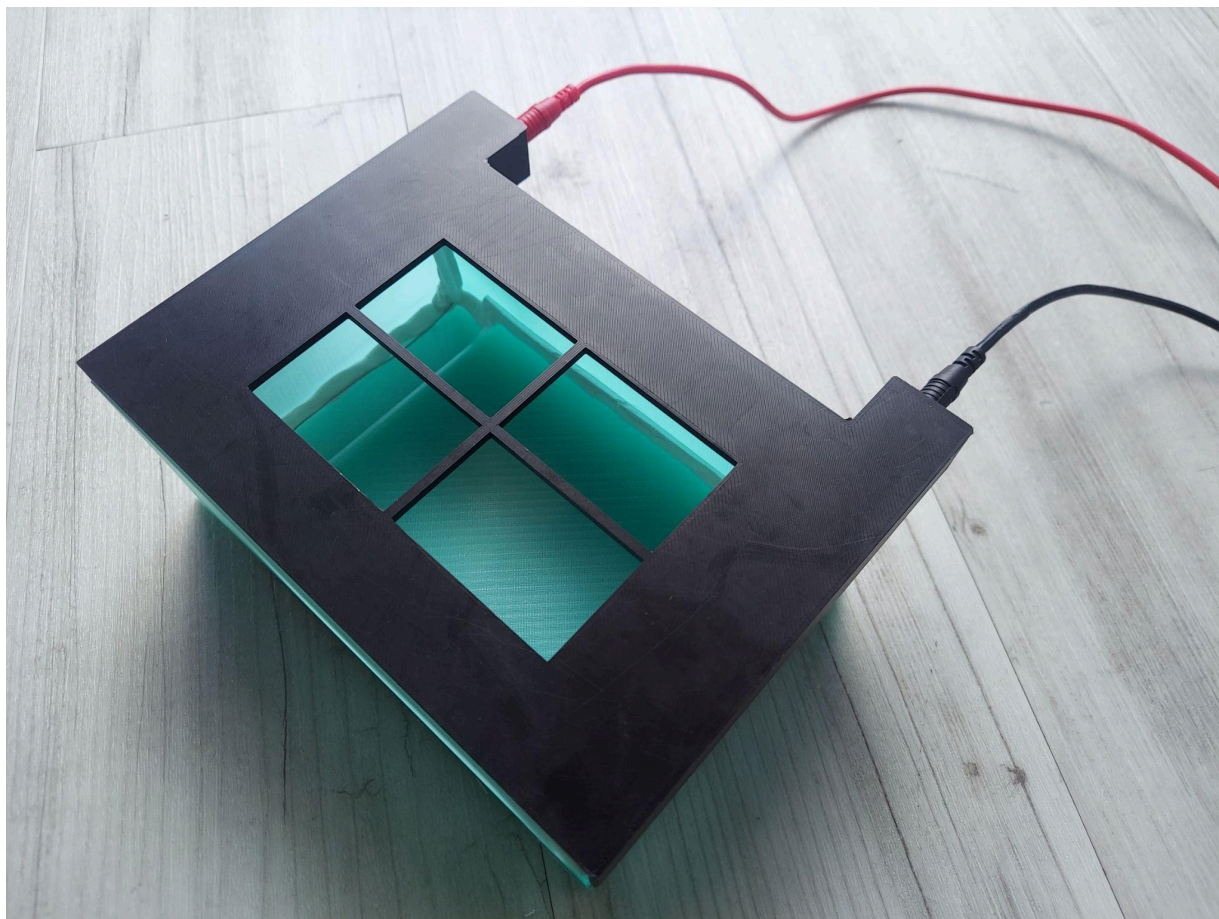


EZ-Phoresis™

Assembly Instructions

Version 1 | September 10, 2026



A compact, 3D-printable gel electrophoresis system designed for research and educational use.

This guide covers printing, component preparation, electrical connector assembly, wire placement, sealing, and final lid installation.

Before you begin

- Review the safety information on the next page.
- Confirm that all printed parts, hardware, wire, tubing, and silicone putty are available.
- Work on a clean, dry surface with the electrophoresis power supply disconnected.

If you used or fabricated the EZ-Phoresis design, please cite the following:

Bastug K. (2026). EZ-Phoresis: a 3D-printable gel electrophoresis system. Version 1.0. NIH 3D.
<https://3d.nih.gov/entries/3DPX-023670>

For the design development and validation study, the associated manuscript citation will be added when available.

Safety Warning and Disclaimer

ELECTRICAL HAZARD

EZ-Phoresis™ is an electrically powered gel electrophoresis device. Electrophoresis power supplies can generate voltages capable of causing electrical shock, serious injury, or death. Users must understand the electrical and laboratory hazards associated with electrophoresis before constructing or operating this device.

Required safety practices

- Always operate the device with the protective lid fully installed.
- Never touch the electrophoresis buffer, electrodes, conductive components, or internal wiring while the system is energized.
- Disconnect the power supply before removing the lid or making any adjustment to the device.
- Inspect the device, electrodes, wiring, connectors, and lid before each use.
- Do not operate a damaged, leaking, improperly assembled, or electrically compromised device.
- Use only a suitable electrophoresis power supply and appropriately rated electrical components.
- Follow applicable institutional laboratory safety procedures and electrical safety requirements.
- Appropriate supervision and training are required.

Disclaimer of liability

These design files, schematics, assembly guides, and associated materials are provided “AS IS,” without warranties of any kind, express or implied.

Users construct, modify, assemble, and operate this device at their own risk and are responsible for determining whether the device is appropriate and safe for their intended application.

To the extent permitted by applicable law, the designer and contributors assume no liability for injury, property damage, data loss, experimental loss, or other damages arising from fabrication, modification, assembly, operation, or use of the device.

EZ-Phoresis™ is intended for research and educational use and has not been validated or approved for clinical diagnostic use.

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Overview & 3D Printing Guidance

This document provides instructions to assemble the EZ-Phoresis™ system. Ensure that you have all supplies listed in the Bill of Materials before beginning. This guide assumes a baseline understanding of 3D printing and gel electrophoresis.

Required tools: scissors, wire cutters, pliers (straight or needle-nose), and a standard kitchen butter knife.

Standard printing specifications (use unless otherwise specified)

- 2–3 perimeters
- 15–20% infill
- No structural supports
- 0.2 mm layer height

STL files for all components are intended to be distributed through the project GitHub repository and the NIH 3D Print Exchange. Add the final links before publication.

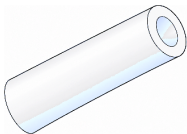
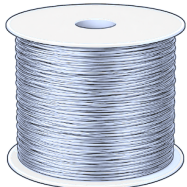
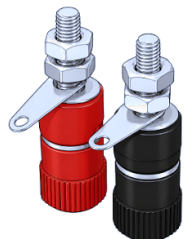
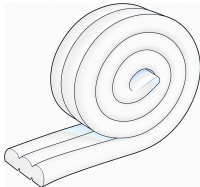
Component	File name	Description	Material	Additional specifications
A	Lid_EZ-Phoresis.stl	Gel electrophoresis lid	PLA	Standard settings
B	Base_EZ-Phoresis.stl	Gel electrophoresis base; requires further assembly	PETG	5 perimeters; ≥90% infill
C	Comb_EZ-Phoresis.stl	Double-sided comb: 15 wells (4 × 1.5 × 8.5 mm) or 11 wells (6 × 1.5 × 8.5 mm)	PLA or PETG	Standard settings
D	Tray_double_EZ-Phoresis.stl	Gel-casting tray with attachment points for up to two combs	PLA or PETG	≥90% infill
E	Tray_triple_EZ-Phoresis.stl	Gel-casting tray with attachment points for up to three combs	PLA or PETG	≥90% infill

Filament selection

- Any color may be used, but a light-colored or translucent filament may make bands easier to see during electrophoresis.
- The system was tested using PETG for the base and PLA for the other components.
- PLA may be sufficient for the base, but may be less durable over time and may have poorer water impermeability.

Materials for Base Assembly

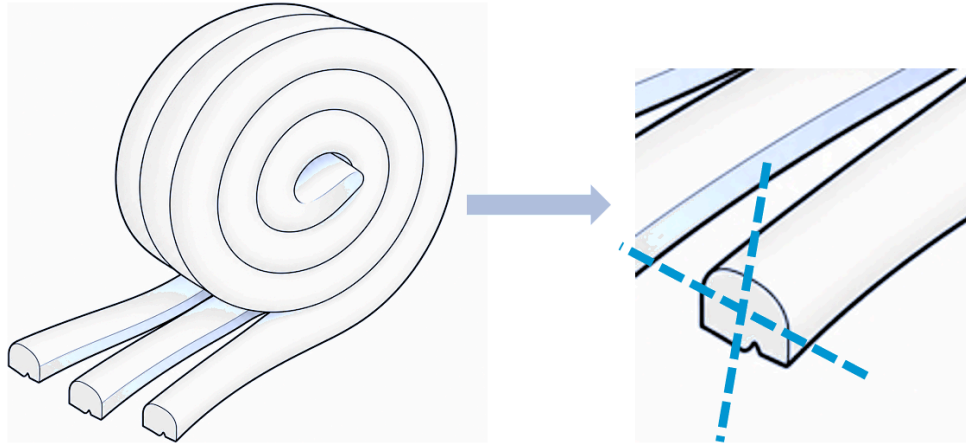
The following materials are required to assemble Component B (the electrophoresis base).

Image	Description	Quantity	Preparation / notes
	PTFE (Teflon) tubing	4 pieces, 4.5 cm each	2 mm ID × 4 mm OD
	22-gauge stainless-steel wire	2 pieces, 25 cm each	See electrode material note below
	Female banana socket, solder-free	1 red and 1 black	Remove the ring terminal; it is not needed
	Moldable silicone putty, white	As needed	Cut into thin strips, approximately ¼ cm wide

Electrode material note. Platinum wire is standard in commercially available electrophoresis systems and may also be used with EZ-Phoresis™. Stainless steel is more widely available and substantially less expensive, but will require more frequent replacement because of anodic dissolution during electrolysis.

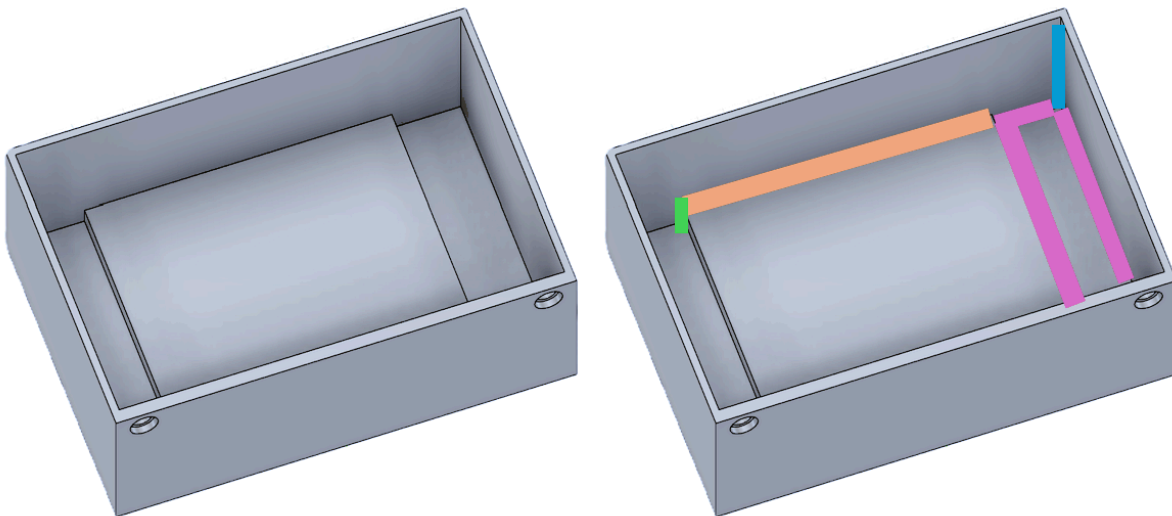
Assembly Instructions — Base

- 1 Prepare the silicone putty.** Peel sections from the roll and divide each section lengthwise into quarters to create thin, long strips.



Cut the silicone putty into narrow, continuous strips.

- 2 Seal the inside corners and seams.** Press silicone putty into the locations shown*. This improves resistance to liquid leakage and can be removed and re-applied as needed. Putty does not dry out when exposed to the air.



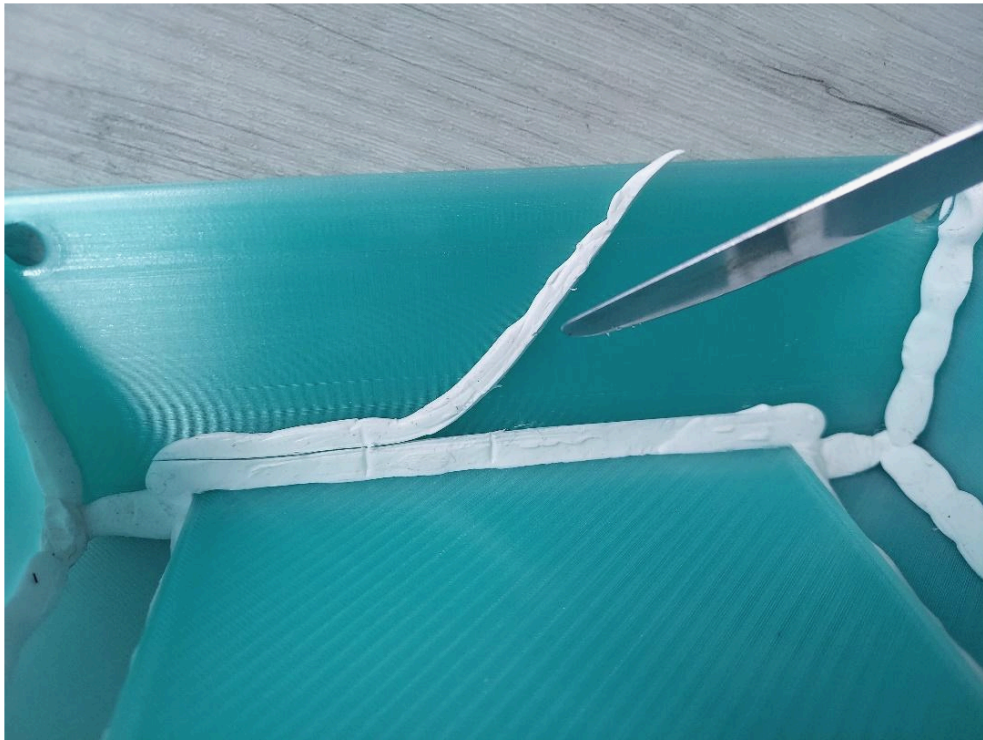
**Apply putty to the four vertical corners, tray-support seams, base perimeter sections, and interior vertical sections.*

- Blue: four inside vertical corners
- Orange: two long tray-support seams
- Pink: four sides on each end (8 strips total)
- Green: four interior vertical sections



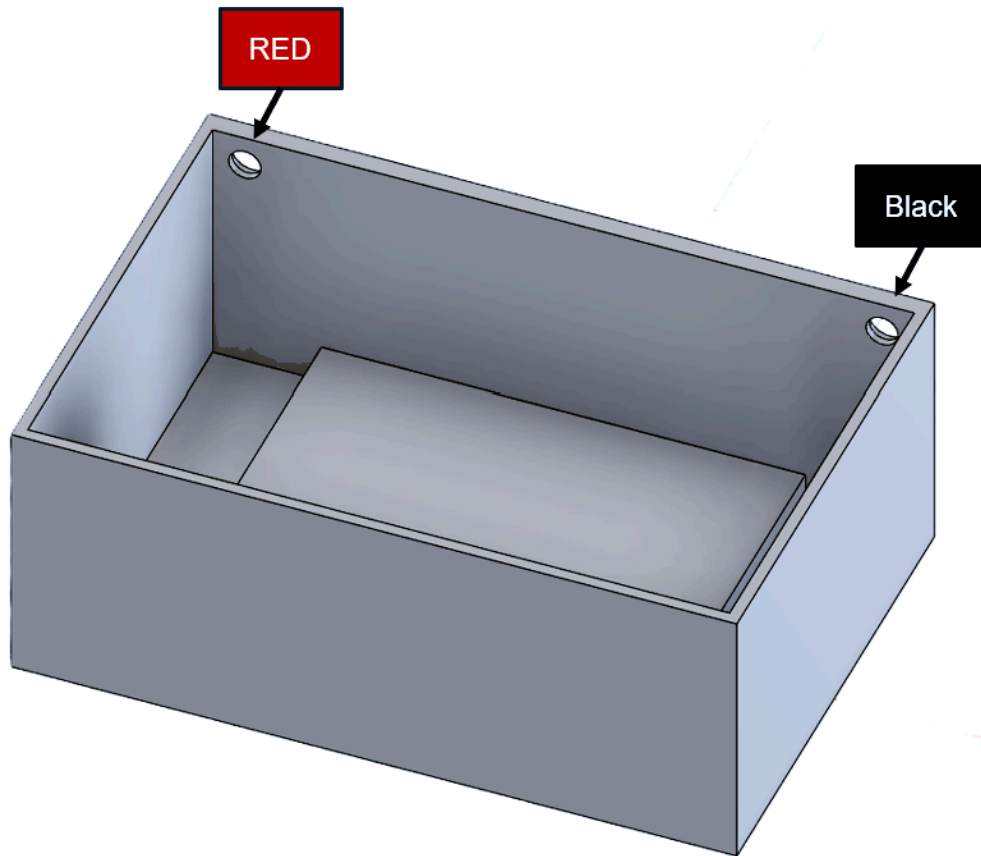
3

Smooth the silicone putty. Use a flat kitchen knife to create neat, continuous edges. This step is optional.



4

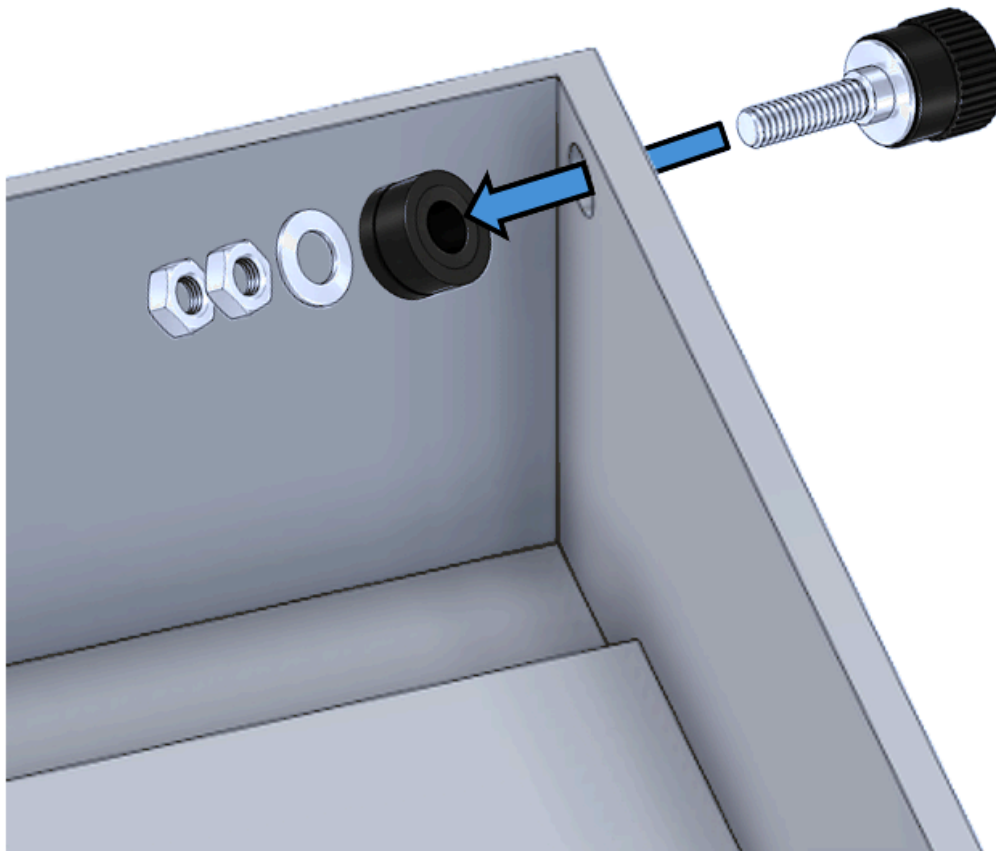
Install the red and black banana sockets. Attach the sockets to the two holes in the base. Use the red socket at one end and the black socket at the opposite end, as shown.



Assembly Instructions — Electrical Connections

5

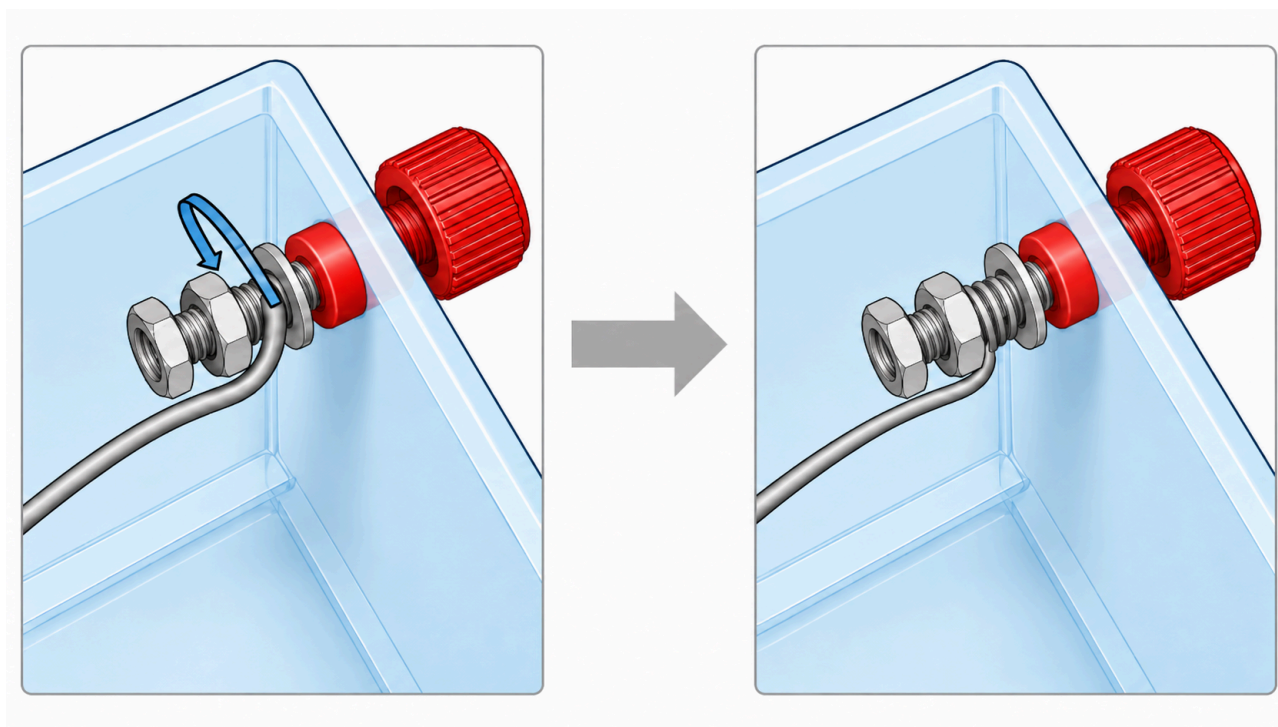
Loosely assemble each banana socket. Insert the socket through the wall and assemble the washer and nuts on the inside. Leave the inner nuts loose enough to receive the stainless-steel wire. The outer red/black connector receives the power-supply lead.



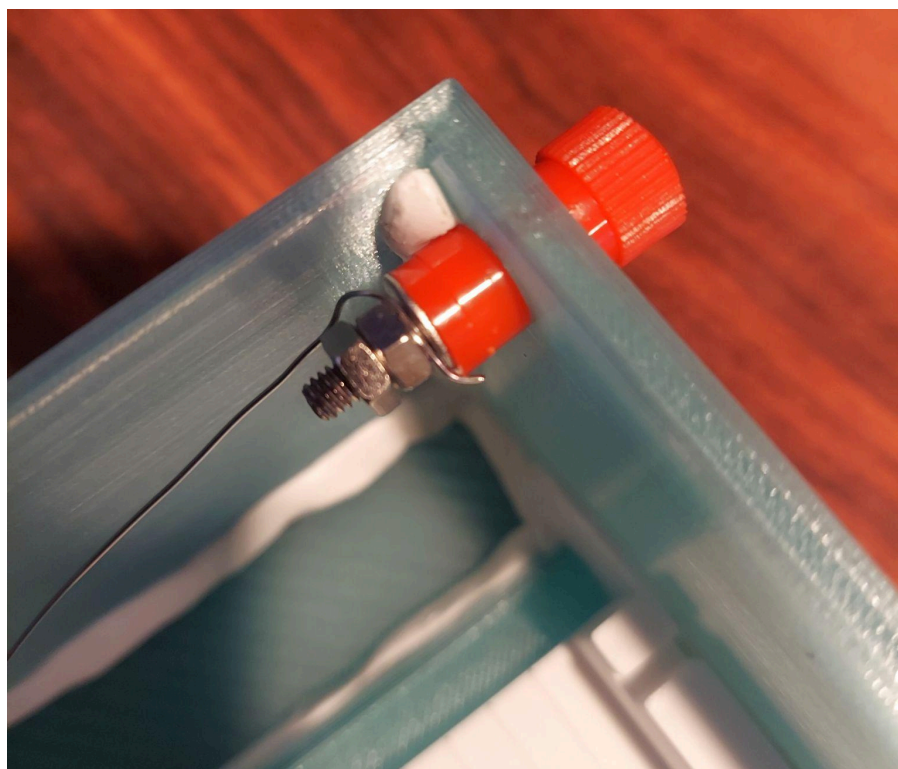
Exploded view of the solder-free banana socket assembly.

6

Wrap the stainless-steel wire around the terminal. Using pliers, wrap the wire 1–2 times around the threaded post between the washer and the first inner nut. Keep the wire snug against the post. Tighten the nuts to secure the wire.



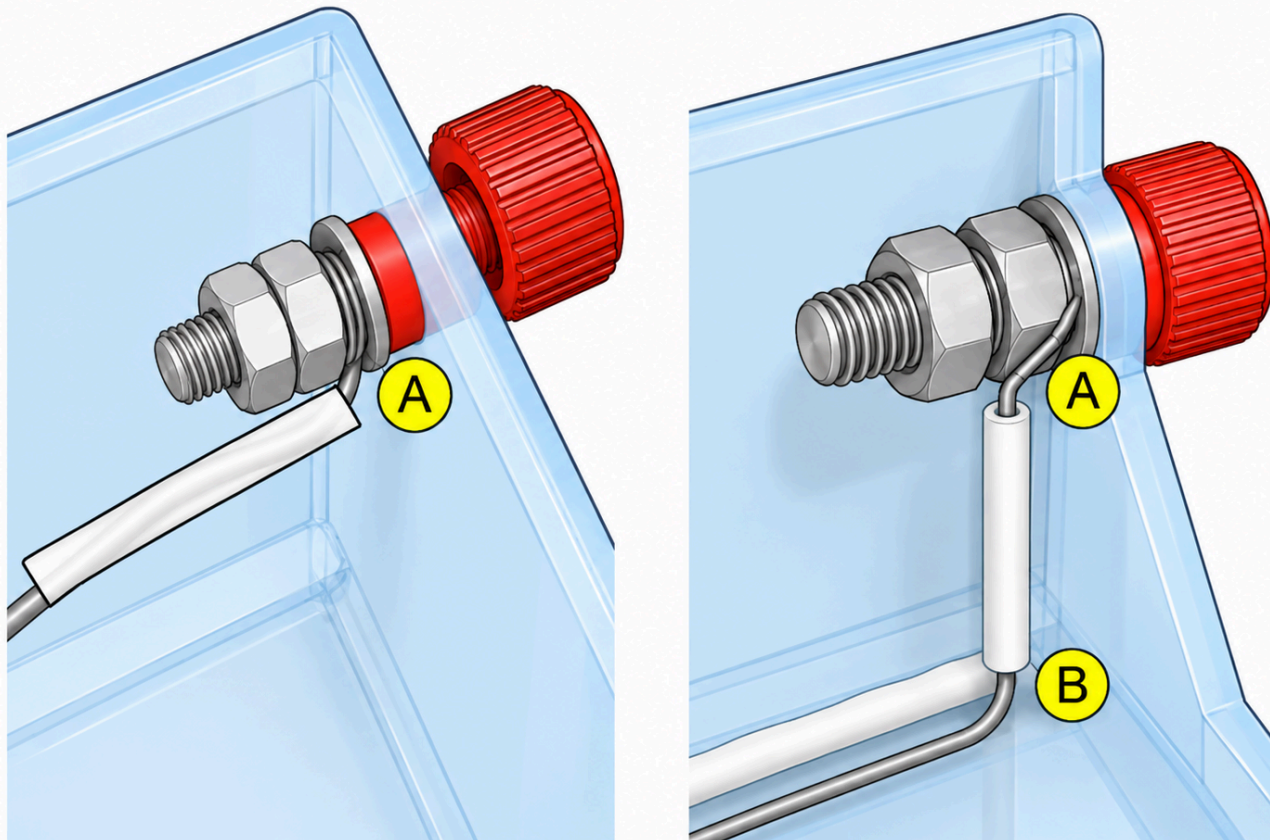
Wrap the wire around the threaded post, then seat the wrap beneath the washer/nut assembly. Excess wire remains in the photograph below to denote the location.



Assembly Instructions — Shape the Electrodes

7

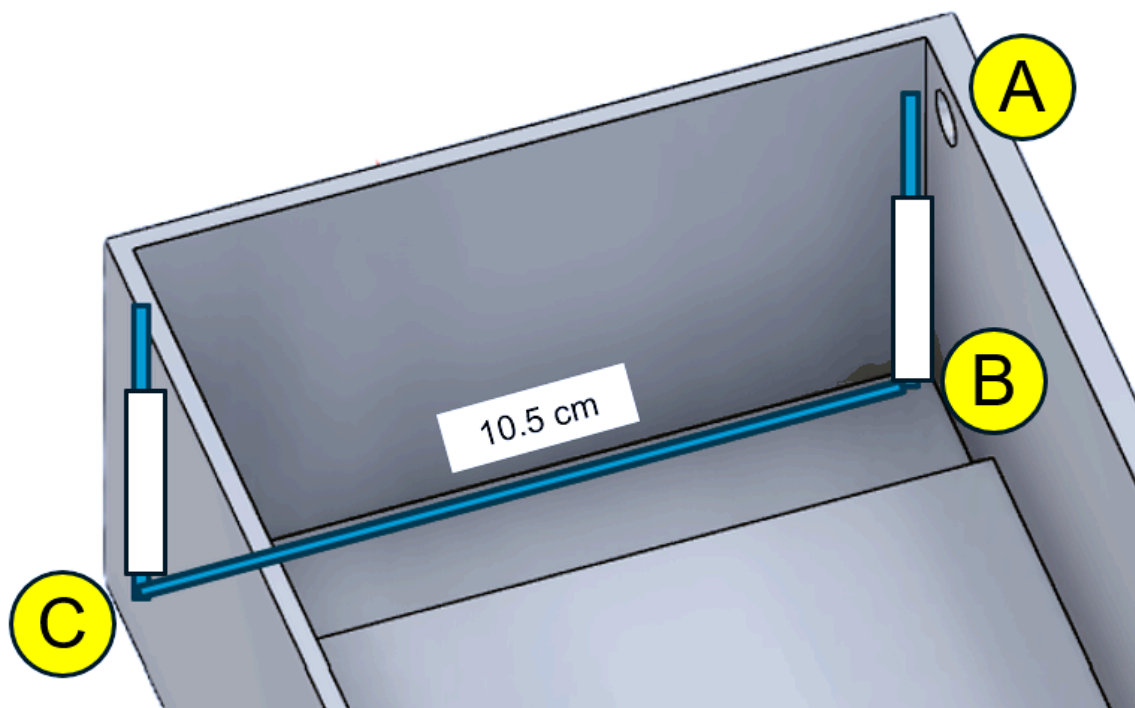
Insulate and make the first bend. Slide one 4.5 cm piece of PTFE tubing onto the wire. At Point B, make a 90° bend so the wire runs parallel to the short edge of the box.



Point A marks the terminal connection; Point B marks the first 90° bend.

8

Measure the electrode span and make the second bend. Measure 10.5 cm from Point B and make a second 90° bend at Point C. Slide another piece of PTFE tubing over the vertical wire. Trim excess wire and fine-tune the length as needed.



Target geometry for the wire between Points A, B, and C. The horizontal span is 10.5 cm.



Assembly Instructions — Secure & Finish

9

Secure the insulated wire. Use light pressure to seat the PTFE tubing into the silicone putty. The exposed stainless-steel wire may also be pressed gently into the putty for stability. Trim any excess wire at the terminal as needed.

Note: It is important that the stainless steel wire is as straight as possible, to ensure a uniform electric field



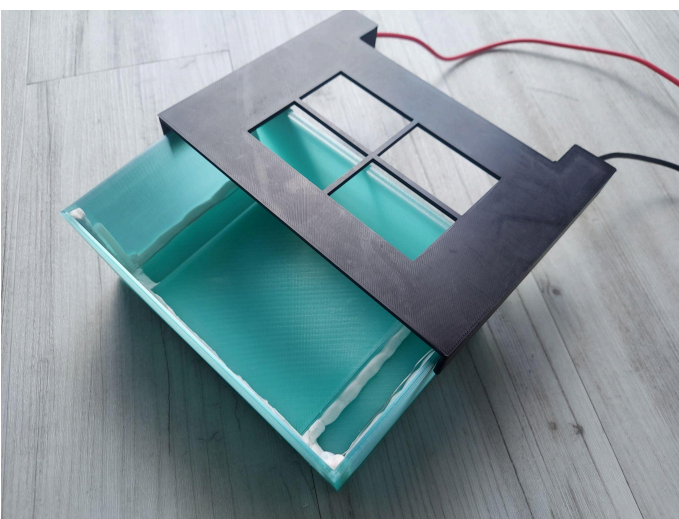
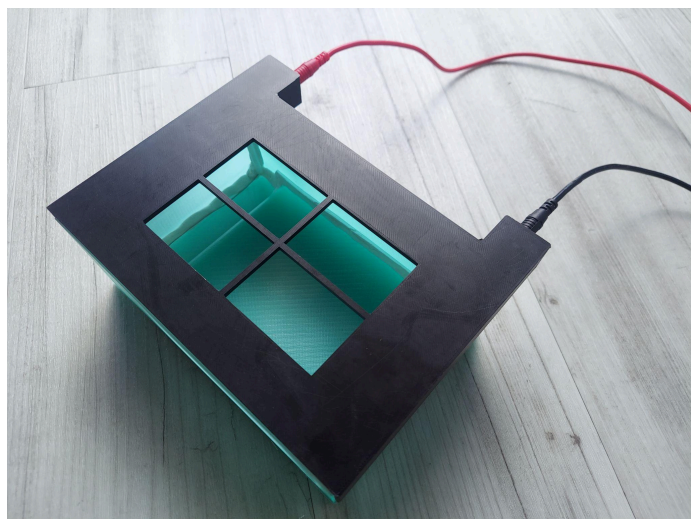
Completed electrode positioned and supported by the silicone putty.

10

Repeat on the opposite side. Repeat Steps 5–9 using the black banana socket to create the second electrode.

11

Install the lid and connect the power leads. Align the lid (Component A) with the electrode openings and slide it into position. Insert the red and black power leads through the lid openings and into the corresponding banana sockets.



Assembly is complete!

